COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 19 — Black-Box & White-Box Testing (Test Design)

Section 9: Practical Exercise — Test Design Suite for CEMS

Rigorous test case derivation using Equivalence Partitioning, Boundary Value Analysis, Decision Tables, and State Transition Testing.

1. Equivalence Partitioning & Boundary Value Analysis (Event Capacity: 1 to 100)

Partition Type	Range of Input Values	Representative Test Value	Expected System Behavior
Valid Partition	1 <= attendees <= 100	50 attendees	Accept registration; increment counter.
Invalid Lower Boundary	attendees < 1 (e.g. 0 or negative)	0 or -1 attendees	Reject input with validation error.
Invalid Upper Boundary	attendees > 100	101 attendees	Reject registration; display "Capacity Exceeded".
BVA Boundary Values	Tested at: 0, 1, 2, 99, 100, 101	0 (Fail), 1 (Pass), 100 (Pass), 101 (Fail)	Strict boundary adherence verified.

2. Decision Table Testing (Registration Eligibility Logic)

Condition / Rule	Rule 1	Rule 2	Rule 3	Rule 4
Condition 1: Event Status is OPEN?	TRUE	TRUE	TRUE	FALSE
Condition 2: Available Capacity > 0?	TRUE	TRUE	FALSE	DON’T CARE
Condition 3: Duplicate Registration?	FALSE	TRUE	FALSE	DON’T CARE
Action: Issue Confirmed Registration	YES	NO	NO	NO
Action: Reject with "Already Registered"	NO	YES	NO	NO
Action: Offer Waitlist Placement	NO	NO	YES	NO
Action: Reject with	NO	NO	NO	YES

"Event Closed"				

3. White-Box Branch Coverage Logic

Code logic under test:

```
if (event.isOpen()) {
   if (event.getSeatsLeft() > 0 && !student.isRegistered(eventId)) {
      return confirmRegistration();
   } else { return handleWaitlistOrDuplicate(); }
} else { return eventClosedError(); }
```

Branch Coverage Requirement: Test Suite must execute: (1) `event.isOpen() == false`; (2) `event.isOpen() == true` AND `seatsLeft > 0` AND `!isRegistered == true`; (3) `event.isOpen() == true` AND `seatsLeft <= 0`; (4) `event.isOpen() == true` AND `isRegistered == true`. Achieving 100% branch coverage guarantees all logical decision paths are verified.

Section 12: Homework — Test Design Applications

Practical test case design for student marks entry and e-commerce order state machines.

- **1. Student Marks Entry Form (Allowed Range: 0 to 100):** Equivalence Classes: Valid Class [0 to 100], Invalid Low [< 0], Invalid High [> 100], Invalid Non-Numeric. Boundary Test Inputs: -1 (Error), 0 (Valid), 1 (Valid), 99 (Valid), 100 (Valid), 101 (Error).
- **2. Decision Table for Authentication:** Factors: Username Valid (T/F), Password Valid (T/F), Account Locked (T/F). Expected Actions: Success (T, T, F); Reject Invalid Credentials (F, *, *); Reject Invalid Password (T, F, F); Display Account Locked Error (*, *, T).
- **3. State Transition Matrix for Online Order:** States: CREATED, PAID, SHIPPED, DELIVERED, CANCELLED. Valid Transitions: CREATED -> PAID -> SHIPPED -> DELIVERED; CREATED -> CANCELLED; PAID -> CANCELLED (Refund). Invalid Transitions Tested: DELIVERED -> CANCELLED (Rejected); CREATED -> SHIPPED (Rejected; bypasses payment).
- **4. Regression Test Selection:** After modifying event cancellation rules, select regression test cases covering: (a) Normal cancellation; (b) Re-registration after cancellation; (c) Concurrency during simultaneous cancel/register; (d) Attendance report generation accuracy.